Introduction to Boolean Algebra

Class 11 Vocational Computer Science

What is Boolean Algebra?

Boolean algebra is the mathematics of digital circuits. Unlike standard algebra which deals with continuous numbers, Boolean algebra only deals with two values: 1 (TRUE / HIGH) and 0 (FALSE / LOW). It forms the foundational logic behind how computers process information.

1. Fundamental Logical Operators

There are three primary logical operators that serve as the building blocks for all digital logic.

A. The AND Operator (Logical Multiplication)

The AND operator outputs TRUE (1) only if **all** inputs are TRUE. It is represented by a dot ($\cdot$).

Equation: $Y = A \cdot B$

Input A	Input B	Output Y
0	0	0
0	1	0
1	0	0
1	1	1

B. The OR Operator (Logical Addition)

The OR operator outputs TRUE (1) if **at least one** input is TRUE. It is represented by a plus ($+$).

Equation: $Y = A + B$

Input A	Input B	Output Y
0	0	0
0	1	1
1	0	1
1	1	1

C. The NOT Operator (Logical Inversion)

The NOT operator simply reverses the input. If the input is 1, the output is 0, and vice versa. It is represented by an apostrophe or bar (A' or $\bar{A}$).

Equation: $Y = A'$

Input A	Output Y
0	1
1	0

2. Key Laws of Boolean Algebra

These laws help in simplifying complex Boolean expressions to build efficient digital circuits.

- **Identity Law:** $A + 0 = A$ | $A \cdot 1 = A$
- **Null (Dominance) Law:** $A + 1 = 1$ | $A \cdot 0 = 0$
- **Idempotent Law:** $A + A = A$ | $A \cdot A = A$
- **Involution (Double Negation):** $(A')' = A$
- **Commutative Law:** $A + B = B + A$ | $A \cdot B = B \cdot A$
- **Complementarity Law:** $A + A' = 1$ | $A \cdot A' = 0$
- **Distributive Law:** $A \cdot (B + C) = (A \cdot B) + (A \cdot C)$ | $A + (B \cdot C) = (A + B) \cdot (A + C)$

3. De Morgan's Theorems

Augustus De Morgan introduced two highly important theorems that are crucial for simplifying logical expressions, especially when dealing with NAND and NOR logic gates.

First Theorem

The complement of a sum is equal to the product of the complements.

$$(A + B)' = A' \cdot B'$$

Second Theorem

The complement of a product is equal to the sum of the complements.

$$(A \cdot B)' = A' + B'$$

4. Practical Application

As a computer science student, you will use these rules to simplify equations before writing software conditions or designing hardware circuits. A simpler equation means fewer logical operations, leading to faster execution and cheaper circuit design!